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1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP

Term	Full Form	Use
HTTP	HyperText Transfer Protocol	Used for transferring web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Secure version of HTTP that encrypts data during transmission.
FTP	File Transfer Protocol	Used for transferring files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used for sending emails from a client to a mail server.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4 uses a 32-bit address and supports about 4.3 billion unique addresses (e.g., 192.168.1.1).

IPv6 uses a 128-bit address and supports a vastly larger number of addresses (e.g., 2001:db8::1).

3. Run nslookup google.com on your system and write down the IP address you get.

Address: 10.231.193.218

Non-authoritative answer:

Name: google.com

Addresses: 2404:6800:4000:100c::8b

2404:6800:4000:100c::71

2404:6800:4000:100c::65

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2404:6800:4000:100c::64

192.178.173.102

192.178.173.138

192.178.173.139

192.178.173.100

192.178.173.113

192.178.173.101

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Code Meaning

200 OK – The request was successful.

301 Moved Permanently – The resource has been permanently moved to a new URL.

403 Forbidden – The server understands the request but denies access.

404 Not Found – The requested resource does not exist on the server.

500 Internal Server Error – The server encountered an unexpected error.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

DNS (Domain Name System) converts domain names into IP addresses. The hierarchy is:

1. Root (.)

- The highest level of the DNS hierarchy.
- Represented by an invisible dot at the end of the domain name.

2. TLD (Top-Level Domain)

- In www.google.com, the TLD is **com**.
- Examples: .com, .org, .net, .edu.

3. Domain

- In www.google.com, **google** is the domain name.
- It identifies the organization or website.

4. Subdomain

- In www.google.com, **www** is the subdomain.
- It points to a specific service or section of the domain.

Root (.)

